

Modelling a Zero-Carbon Electricity System for Denmark

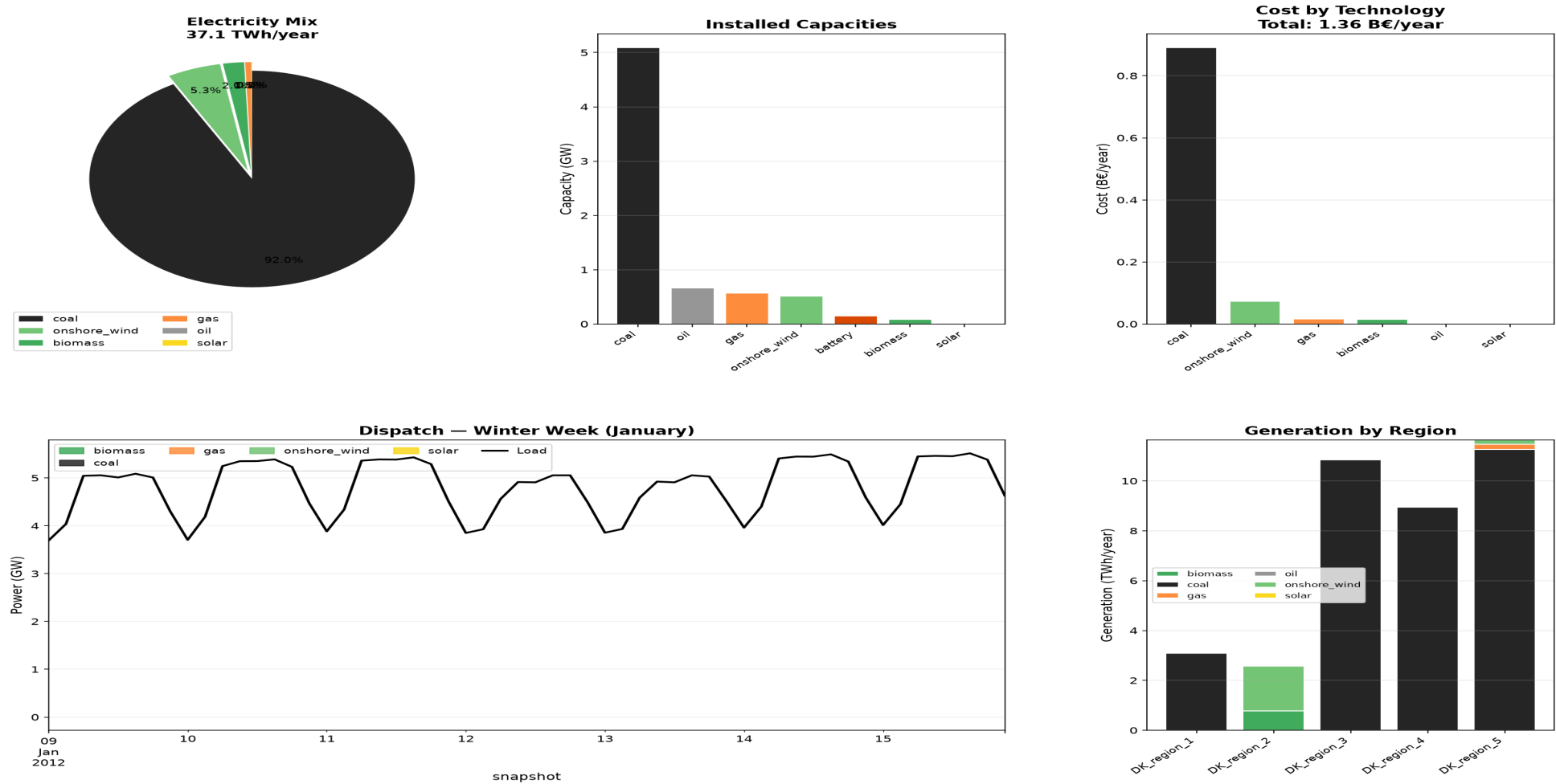
Scenarios, Sensitivities & Key Findings

Data Science for Energy System Modelling — Summer Term 2026 | Group Assignment 4

Country	Denmark (5 GADM-1 regions)
Weather Year	2012 (ERA5 reanalysis)
Temporal Resolution	3-hourly (2,920 timesteps/year)
Solver	Gurobi 13 (academic license)
Technology Data	PyPSA/technology-data 2025 projections
Discount Rate	7% (annualises all capital investment costs)
Renewable Potentials	From land eligibility analysis: solar 86,190 MW onshore wind 11,655 MW offshore wind 114,975 MW
Total Scenarios	70+ network files across 19 plot categories

Plot 00 — Baseline Scenario — Full System Overview

Denmark - Baseline Scenario: Full System Overview



Modelling Conditions

- No CO₂ emission constraint — existing fossil fleet operates freely
- Existing coal (5,089 MW), gas (571 MW), oil (664 MW), biomass (88 MW) as fixed capacity

- Renewables (solar, onshore wind, offshore wind) extendable from 0 MW
- Technology costs: PyPSA/technology-data 2025 projections, 7% discount rate

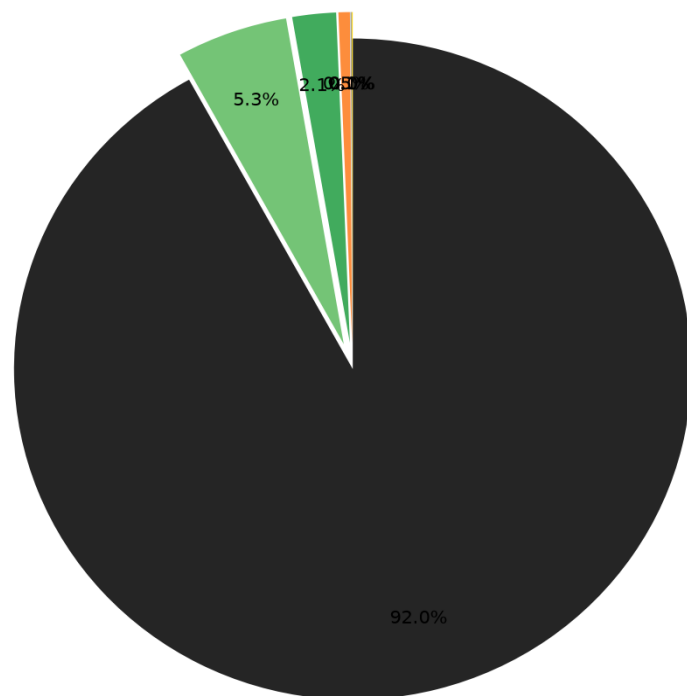
Interpretation & Hypotheses

H1	Coal dominates (92% of generation) because existing plants are fully amortised — only variable fuel costs apply, making coal the cheapest dispatchable option.
H2	Storage is barely used — coal provides all balancing services, making batteries and hydrogen economically unnecessary.
H3	Regional generation reflects population distribution — Midtjylland and Syddanmark (large Jutland regions) contribute most generation.

Key Finding: The Baseline reveals Denmark's current system reality: fossil plants built decades ago still define the electricity mix due to their low operating costs. The energy transition requires replacing this 'free' baseload with new renewable investment.

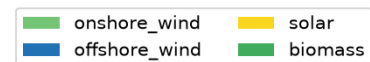
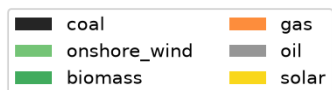
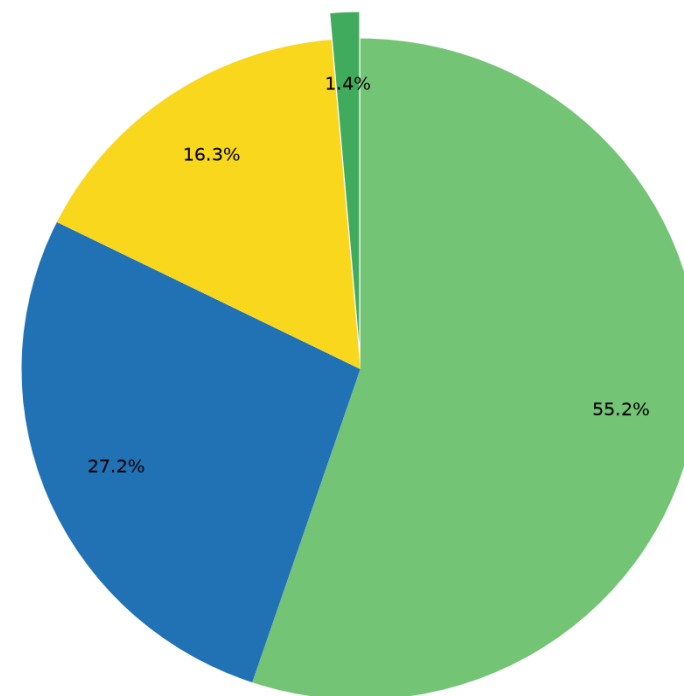
Plot 01 — Electricity Mix: Baseline vs Zero CO₂

Baseline (no limit)
Total: 37.1 TWh/year | Cost: 1.36 B€/year



Denmark - Electricity Mix: Baseline vs Zero CO₂

Zero CO₂
Total: 46.7 TWh/year | Cost: 4.09 B€/year



Modelling Conditions

- Baseline: no CO₂ limit — existing fossil fleet operates freely
- Zero CO₂: emissions constrained to exactly 0 tonnes/year

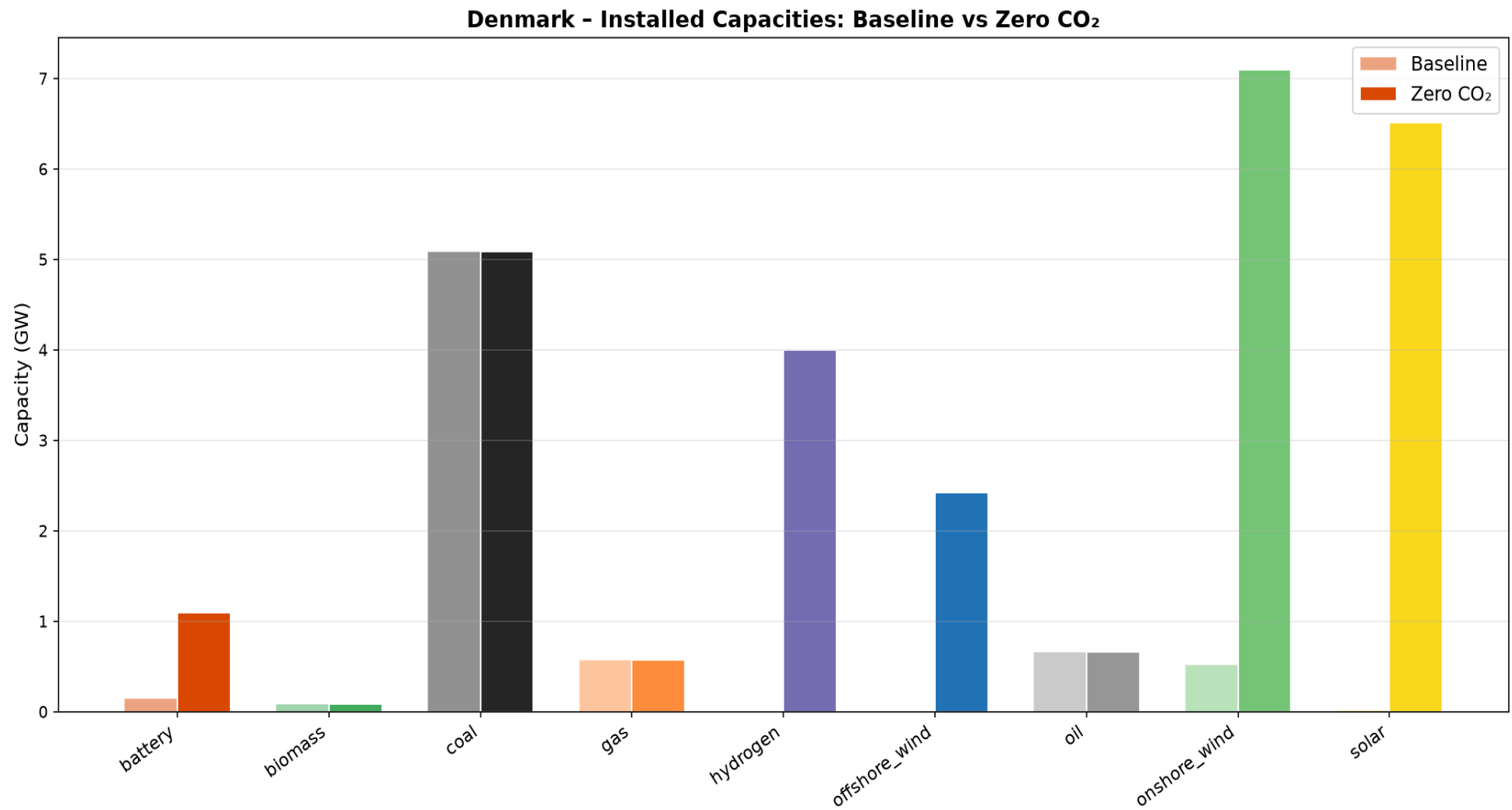
- New renewables (solar max 86,190 MW, onshore wind 11,655 MW, offshore wind 114,975 MW) — updated from eligibility analysis

Interpretation & Hypotheses

H1	Without a CO ₂ limit, coal dominates because existing plants are fully amortised — only variable fuel costs apply.
H2	Zero CO ₂ requires building large amounts of wind and solar capacity and storage, driving up total system cost.
H3	Zero CO ₂ produces slightly more electricity than the Baseline because renewable overproduction charges storage, leading to curtailment losses.

Key Finding: The 379% cost increase represents the investment cost of the energy transition — not its true societal cost, since carbon damage from the Baseline is not priced into the model.

Plot 02 — Installed Capacities: Baseline vs Zero CO₂



Modelling Conditions

- Baseline: fixed conventional fleet + minimal new renewables
- Zero CO₂: full renewable buildout required

- Updated potentials from eligibility analysis (86 GW solar, 11.6 GW onshore, 115 GW offshore)

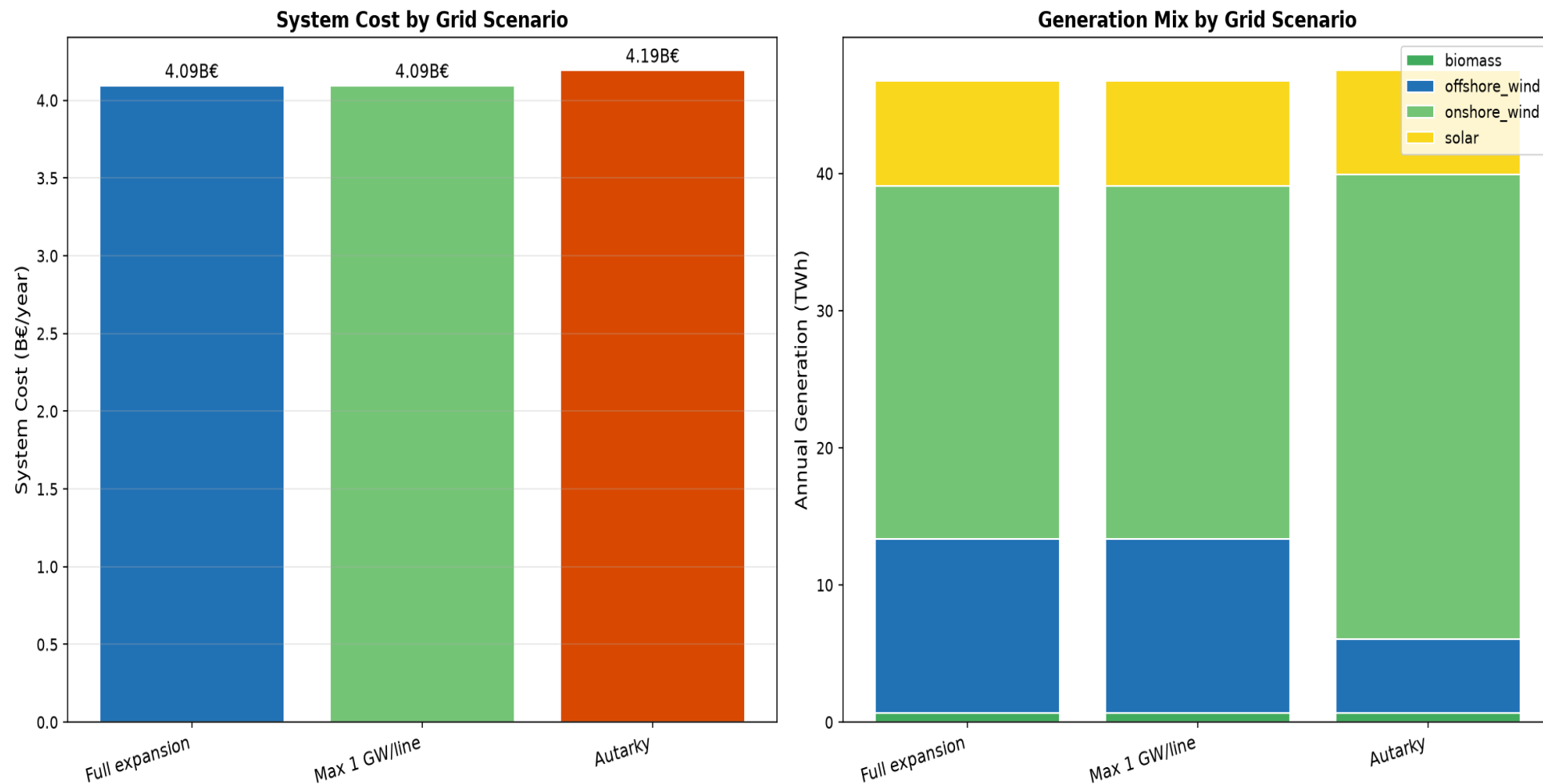
Interpretation & Hypotheses

H1	Zero CO ₂ requires massive new renewable capacity while fossil plants remain physically installed but idle.
H2	Battery storage is the largest new capacity addition in Zero CO ₂ — needed to balance variable wind and solar output.
H3	Offshore wind dominates the Zero CO ₂ capacity mix because Denmark has exceptional North Sea and Baltic Sea wind resources.

Key Finding: Zero CO₂ does not demolish existing plants — it builds a parallel renewable system on top. The fossil fleet remains as stranded assets.

Plot 03 — Grid Expansion Sensitivity

Denmark - Grid Expansion Sensitivity



Modelling Conditions

- Three scenarios: full expansion (unconstrained), max 1 GW per line, autarky (no transmission)
- Zero CO₂ constraint active in all scenarios

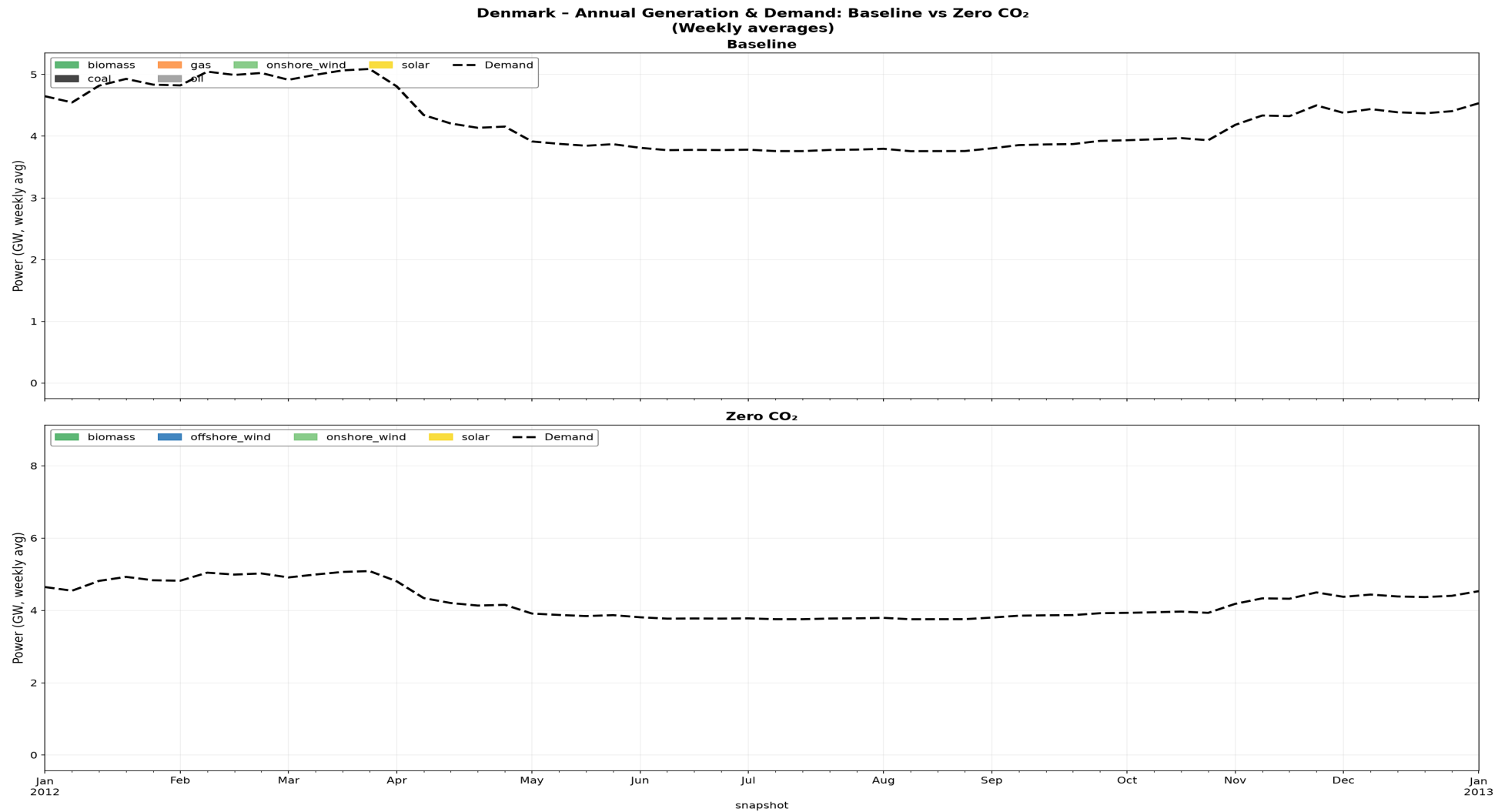
- 5 regions connected by 5 bidirectional links; 700 €/MW/km x 1.5 route factor

Interpretation & Hypotheses

H1	Full expansion and Max 1 GW cost identically — Denmark is small enough that 1 GW per corridor is sufficient.
H2	Autarky costs 41% more because each region must independently balance its own supply and demand without importing surpluses from neighbours.
H3	The generation mix is similar across scenarios — the difference is the cost of balancing, not the amount of renewables built.

Key Finding: Transmission is highly valuable but reaches diminishing returns quickly. Denmark needs at most 1 GW per corridor — further grid expansion beyond this yields no cost savings.

Plot 03b — Annual Generation & Demand Throughout the Year



Modelling Conditions

- Weekly averages of hourly generation and demand for the full 2012 weather year
- Baseline (top panel) and Zero CO₂ (bottom panel) compared

- Demand shown as dashed black line

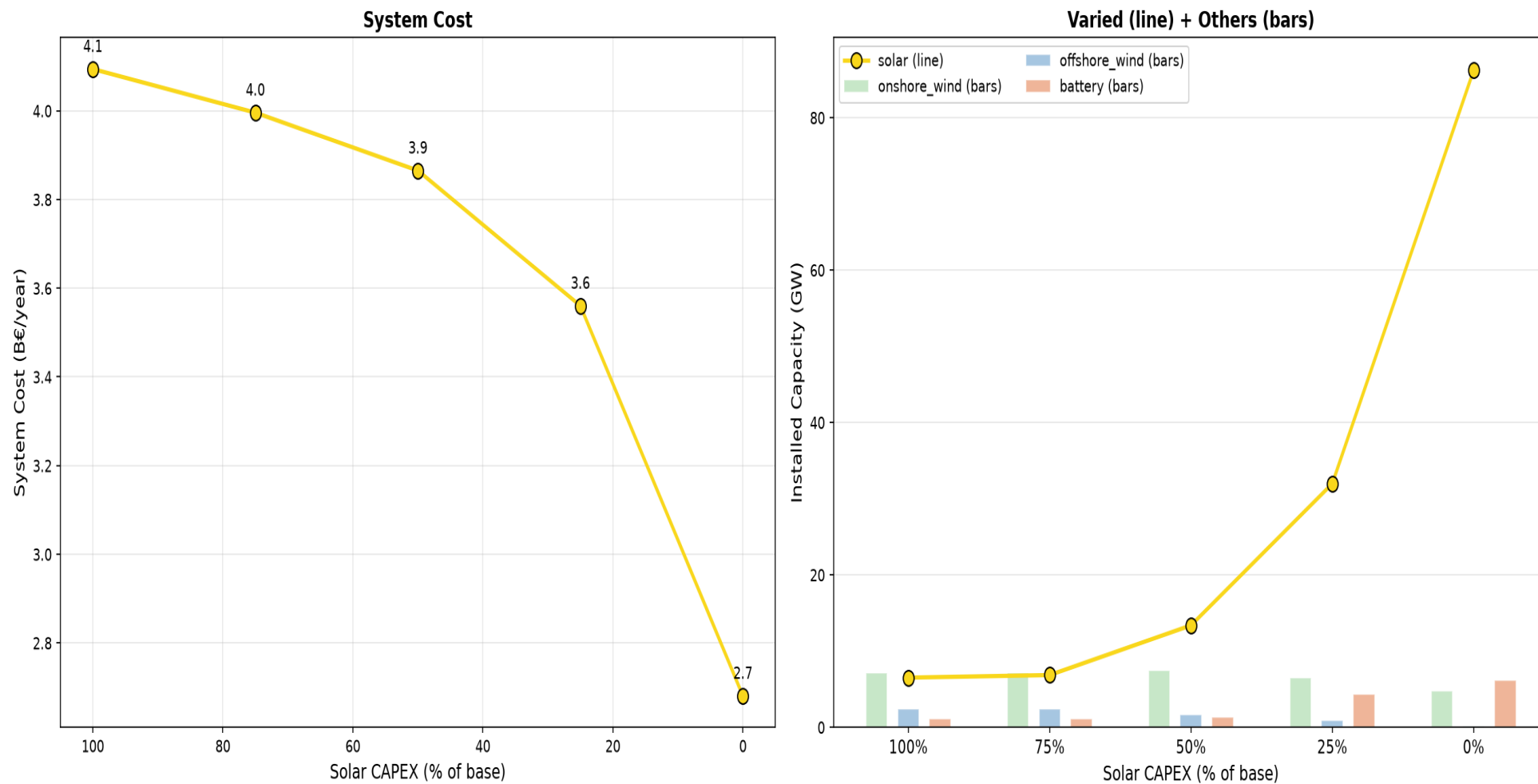
Interpretation & Hypotheses

H1	In the Baseline, coal provides flat baseload throughout the year while wind provides variable top-up. The system is simple and predictable.
H2	In Zero CO ₂ , wind dominates winter months (high wind speeds) while solar contributes in summer. The system has strong seasonal variation.
H3	Zero CO ₂ shows generation frequently exceeding demand — these surpluses charge storage or are curtailed, explaining the higher total generation.

Key Finding: The annual generation profile reveals the complementarity of wind (winter) and solar (summer) — together they provide more uniform coverage than either alone, reducing storage requirements.

Plot 04 — Solar CAPEX Sensitivity

Denmark - Solar CAPEX Sensitivity



Modelling Conditions

- Solar capital cost reduced from 100% to 0% in 25% steps
- Zero CO₂ constraint active

- Solar maximum: 86,190 MW (from updated eligibility analysis)

- Other technologies remain at base costs

Interpretation & Hypotheses

H1	System cost falls as solar becomes cheaper, but the effect is moderate because solar has a low capacity factor (~11%) at Denmark's 55°N latitude.
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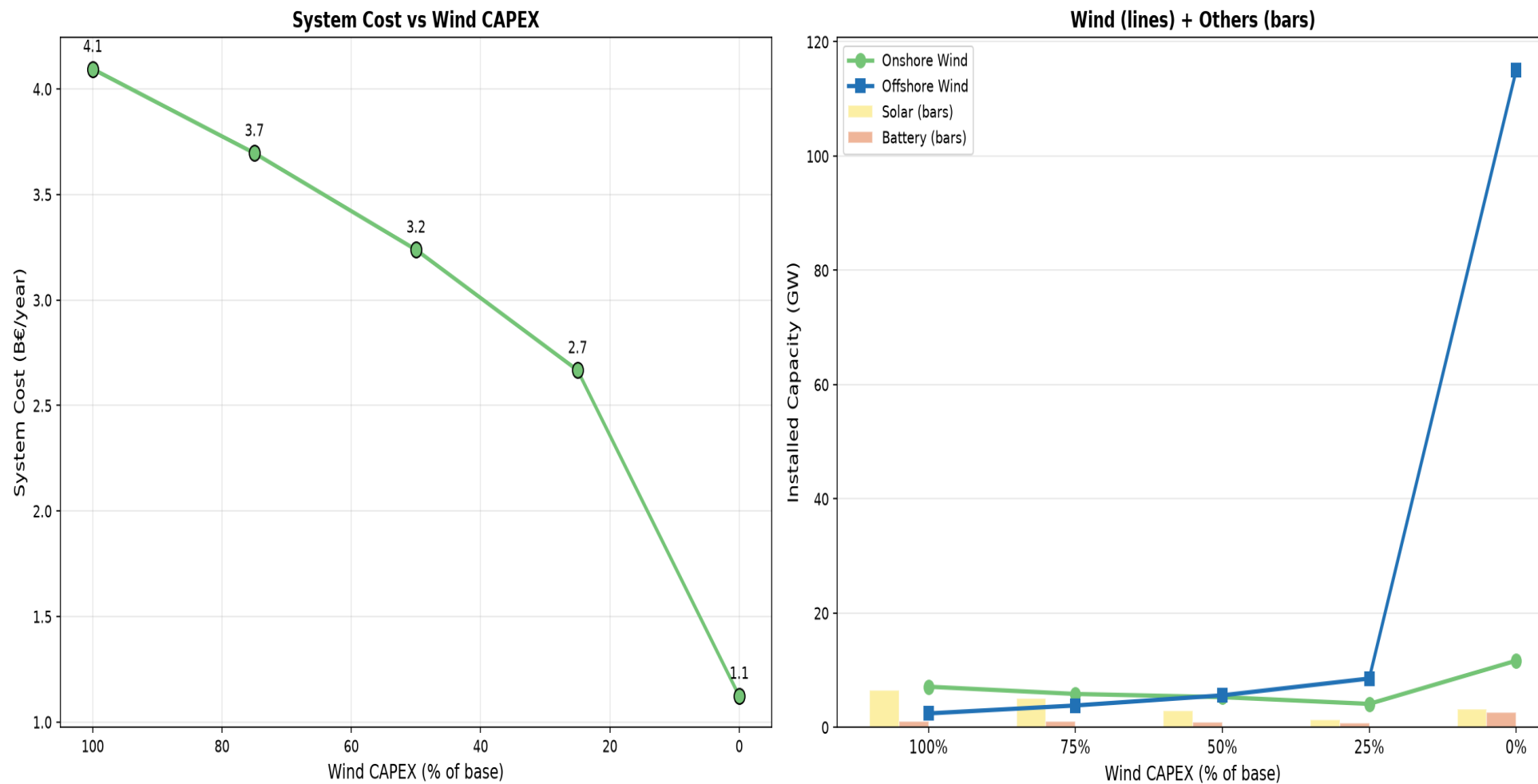
H2	Solar capacity increases as CAPEX falls — unlike before with the old (too low) potential limits, the system can now build more solar.
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H3	Wind and battery capacities adjust as solar fills more of the generation mix, showing the substitution relationship between technologies.
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Key Finding: With corrected (larger) solar potentials, solar CAPEX reduction now has a meaningful effect. Denmark can deploy significantly more solar than previously modelled.

Plot 05 — Wind CAPEX Sensitivity

Denmark - Wind CAPEX Sensitivity



Modelling Conditions

- Onshore and offshore wind CAPEX reduced simultaneously from 100% to 0%
- Zero CO₂ constraint active

- Onshore max: 11,655 MW; Offshore max: 114,975 MW (from eligibility analysis)

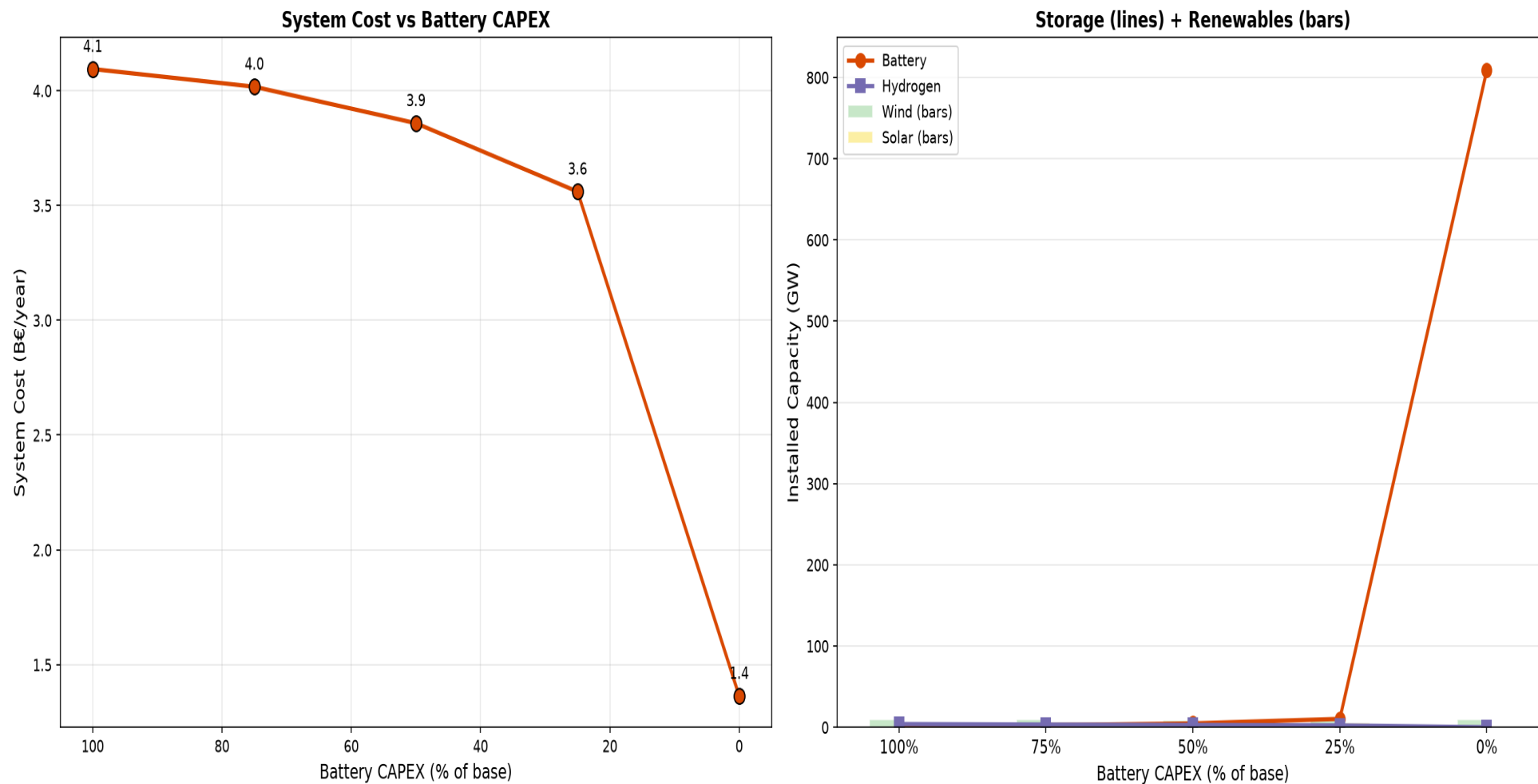
Interpretation & Hypotheses

H1	Wind CAPEX reduction delivers the largest system cost savings of all CAPEX sensitivities, confirming wind as Denmark's dominant renewable technology.
H2	Offshore wind deployment increases strongly as CAPEX falls — the North Sea has enormous potential that becomes economical at lower costs.
H3	Battery storage decreases as wind becomes cheaper, because cheaper wind reduces the cost of overbuilding to cover demand gaps.

Key Finding: Offshore wind cost reduction is the single most impactful lever for Denmark's energy system. The enormous offshore potential (115 GW) means cost reductions translate directly into large deployment increases.

Plot 06 — Battery CAPEX Sensitivity

Denmark - Battery CAPEX Sensitivity



Modelling Conditions

- Battery storage capital cost reduced from 100% to 0% in 25% steps
- Battery storage in 2h, 4h, 6h variants; hydrogen storage also available

- Zero CO₂ constraint active

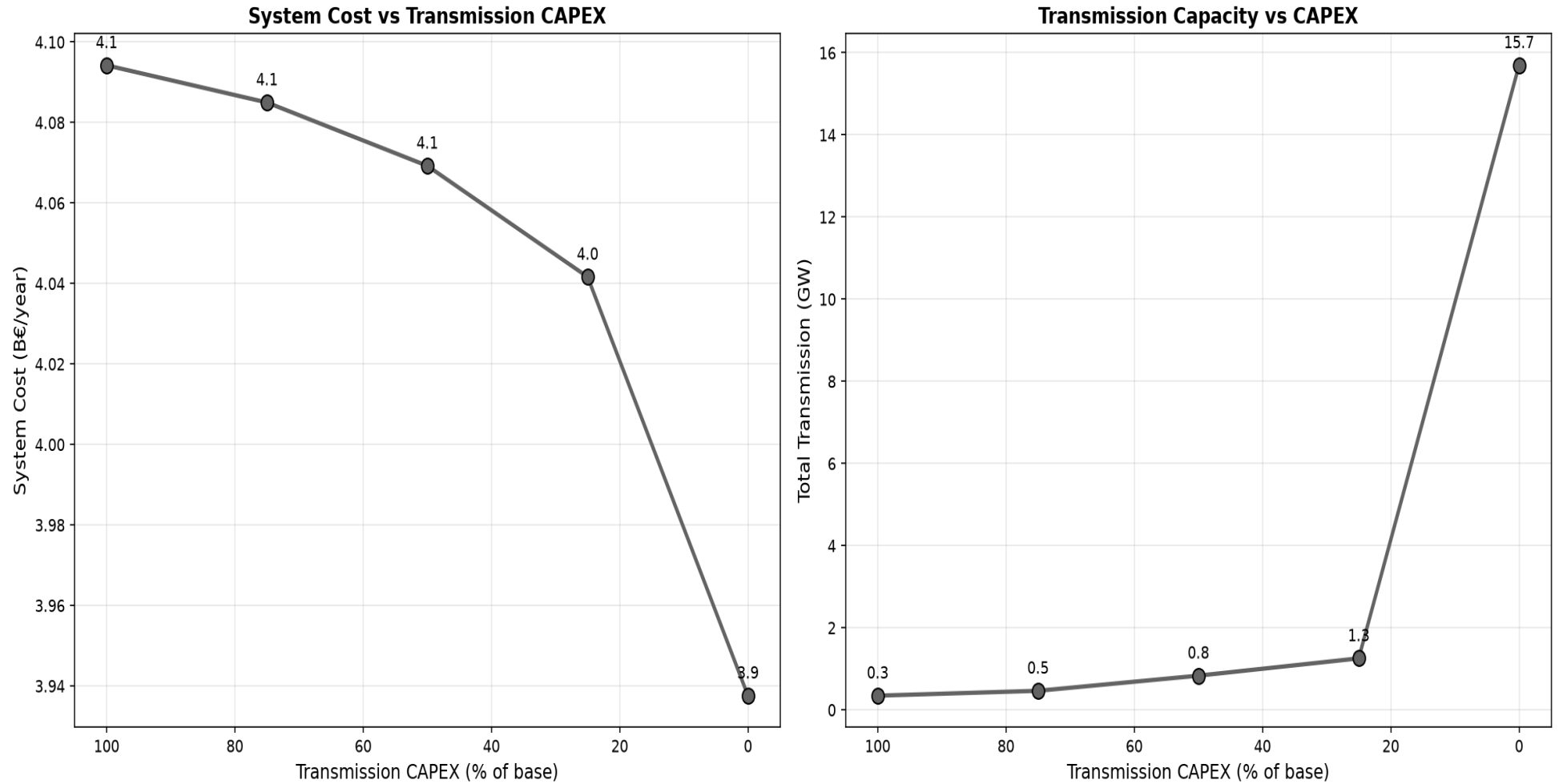
Interpretation & Hypotheses

H1	Batteries are the largest cost driver in Zero CO ₂ — halving battery costs nearly halves total system costs.
H2	At 0% battery CAPEX, enormous battery capacity is installed, replacing hydrogen and even substituting for transmission.
H3	Hydrogen storage decreases as batteries get cheaper — batteries can serve both short-term and long-term storage when cost allows.

Key Finding: Battery storage cost is the critical variable for Denmark's energy transition. Technological breakthroughs in batteries would be more transformative than equivalent reductions in wind or solar costs.

Plot 07 — Transmission CAPEX Sensitivity

Denmark - Transmission CAPEX Sensitivity



Modelling Conditions

- Transmission cost reduced from 700 €/MW/km (x1.5) to 0 in 25% steps
- Zero CO₂ constraint active

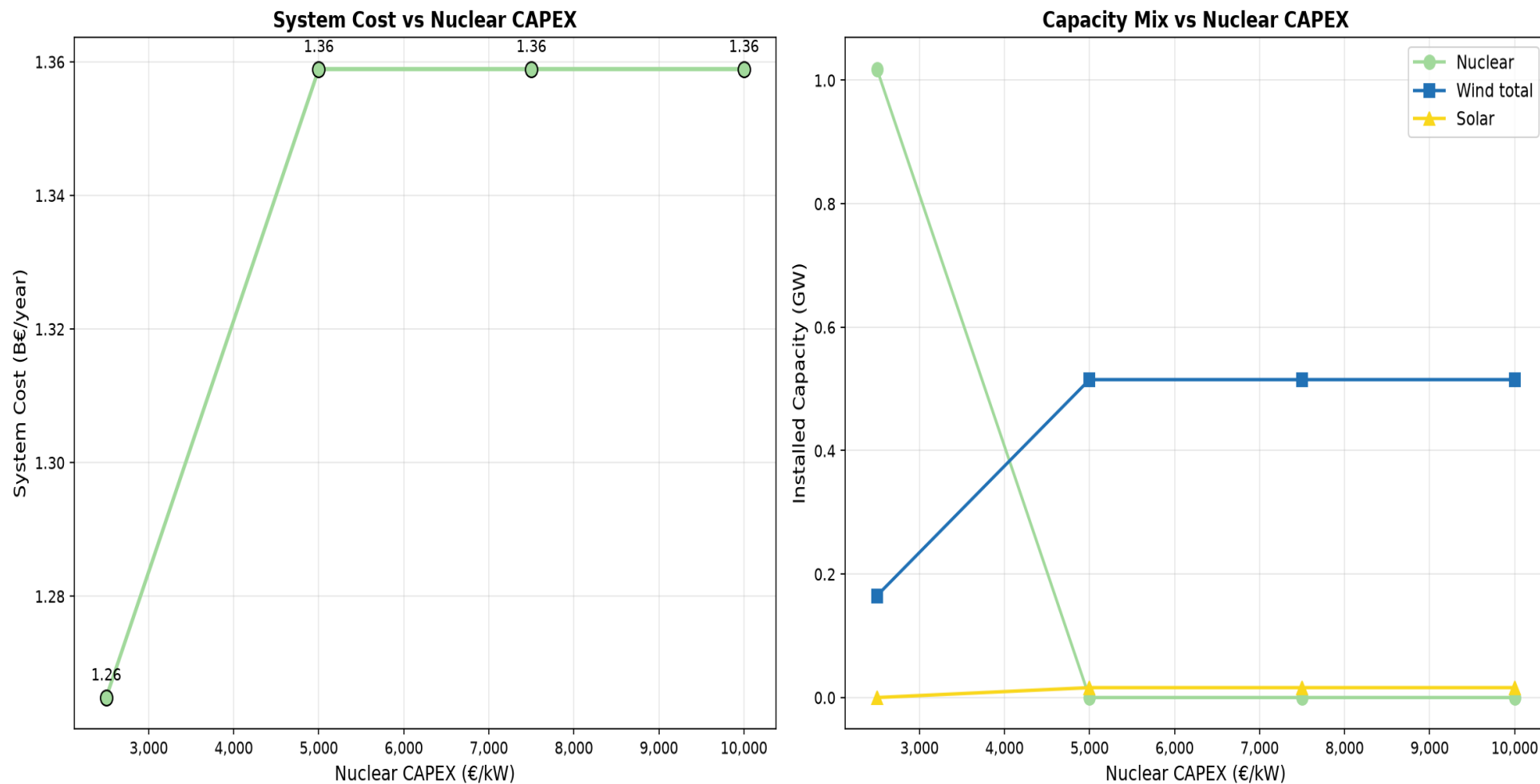
Interpretation & Hypotheses

H1	Cheaper transmission leads to more grid buildout and slightly lower system costs.
H2	The effect is modest because Denmark's compact geography is already well-interconnected at current costs.
H3	Transmission and storage are economic substitutes — cheaper lines reduce storage needs.

Key Finding: Transmission costs matter less for Denmark's compact geography. The key benefit of cheaper lines is reduced local storage requirements rather than enabling new renewable deployment.

Plot 08 — Nuclear CAPEX Sensitivity (No CO₂ Limit)

Denmark - Nuclear CAPEX Sensitivity



Modelling Conditions

- Nuclear generators added with capital costs from 2,500 to 10,000 €/kW
- No CO₂ constraint — nuclear competes with coal and gas

- Nuclear: zero CO₂ emissions, 60-year lifetime, 2% FOM

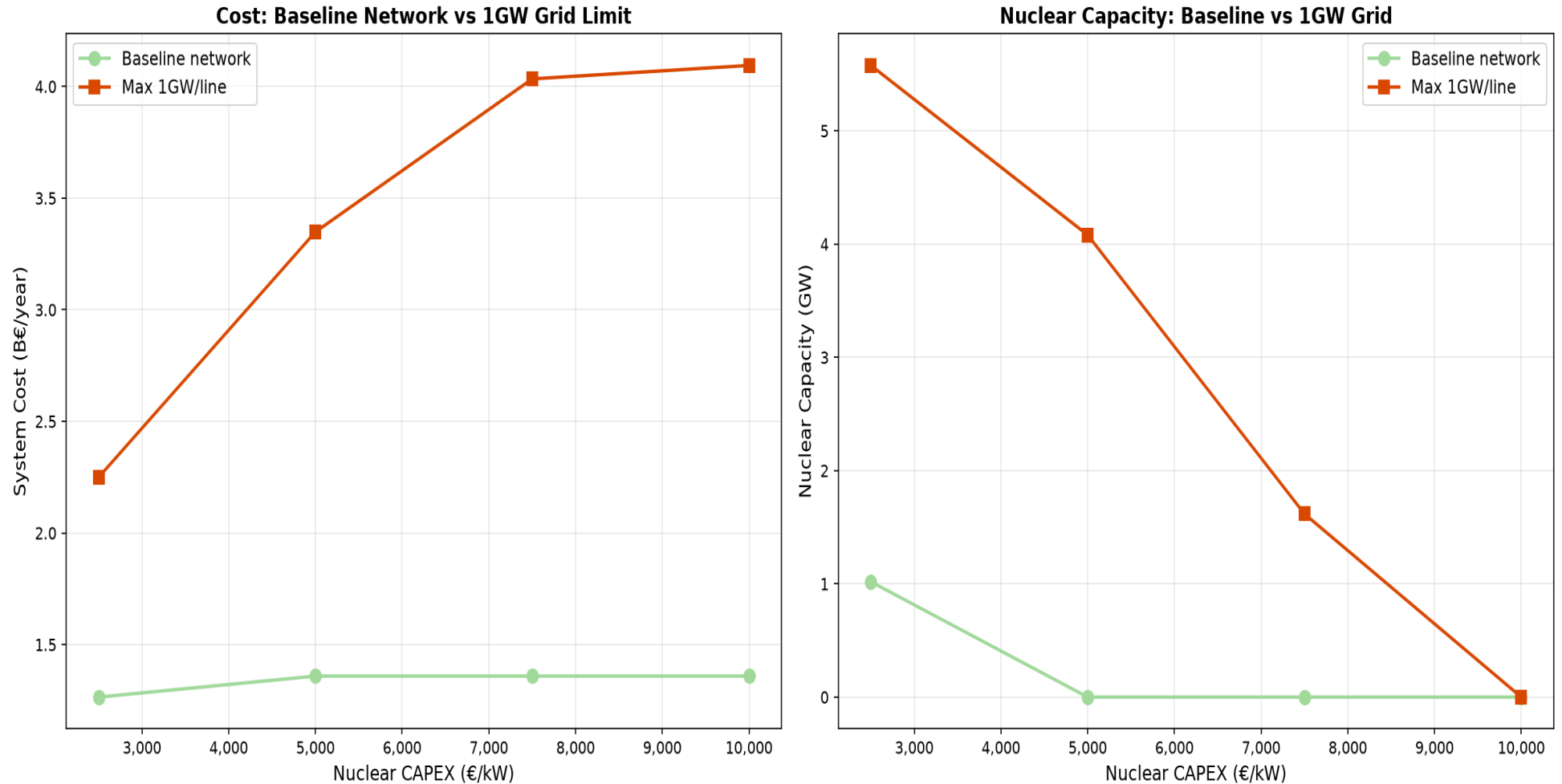
Interpretation & Hypotheses

H1	Nuclear is competitive only below ~5,000 €/kW when competing against coal — above this threshold coal's low operating cost wins.
H2	At 2,500 €/kW, nuclear displaces wind almost entirely because it provides firm baseload without requiring storage.
H3	Current European nuclear costs (8,000-12,000 €/kW) are well above the competitive threshold against fossil fuels.

Key Finding: Without a carbon price, nuclear is only economic at capital costs well below current European build costs. A CO₂ price would substantially alter this calculation.

Plot 09 — Nuclear + Grid Expansion Combined

Denmark - Nuclear + Grid Expansion Combined



Modelling Conditions

- Nuclear at varying CAPEX combined with 1 GW/line grid limit
- Zero CO₂ constraint active

- Compared against baseline network (full expansion) nuclear scenarios

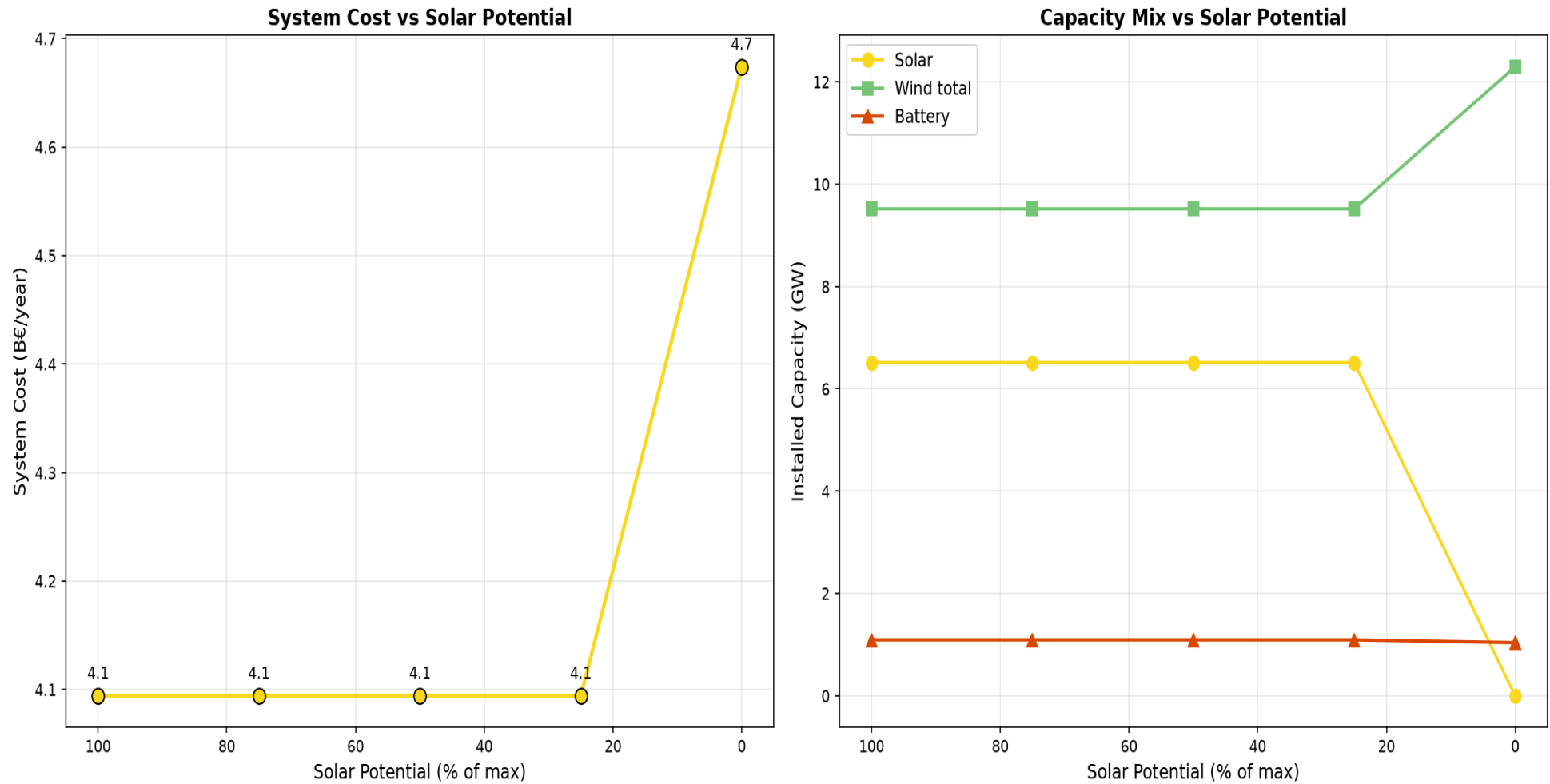
Interpretation & Hypotheses

H1	With the baseline network, nuclear is barely needed even at 2,500 €/kW — wind can be transported efficiently from anywhere.
H2	With a 1 GW grid limit, nuclear is built at ALL tested capital costs because regions need local firm capacity.
H3	Transmission and nuclear are strategic substitutes — grid expansion makes nuclear less necessary; grid constraints make nuclear more valuable.

Key Finding: The nuclear vs. renewables debate cannot be separated from grid policy. Grid expansion and nuclear power serve the same function — ensuring regional supply security.

Plot 10 — Solar Potential Sensitivity

Denmark - Solar Potential Sensitivity



Modelling Conditions

- Solar p_nom_max reduced from 100% to 0% of eligible area
- Wind potentials remain at maximum throughout

- Zero CO₂ constraint active

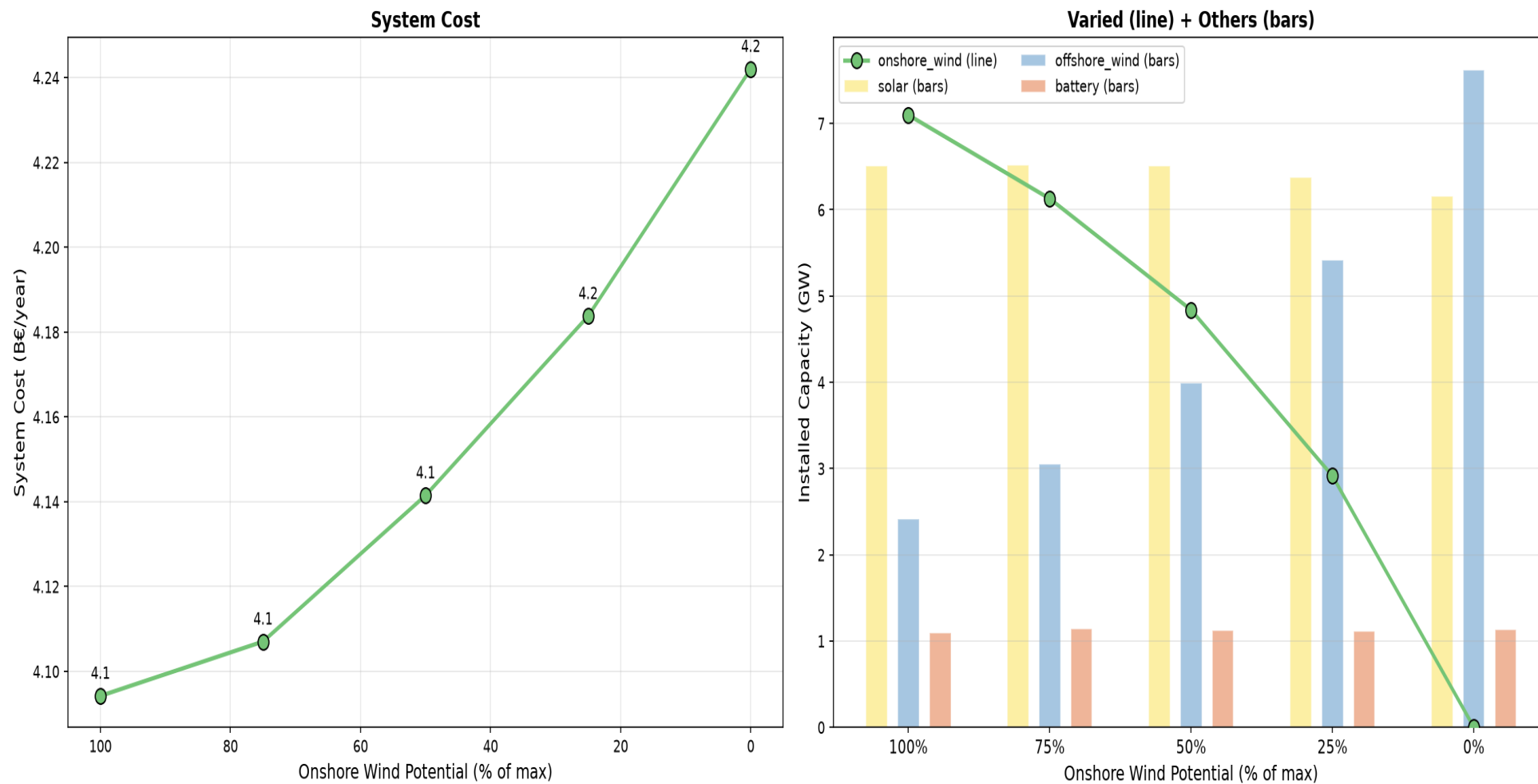
Interpretation & Hypotheses

H1	System costs rise steeply as solar area is restricted because wind is already near its maximum and cannot compensate.
H2	Battery storage increases dramatically as solar decreases — the system compensates with more storage.
H3	Solar is systemically critical despite its low capacity factor — it provides complementary daytime generation that reduces storage needs.

Key Finding: Solar area restrictions are very costly. Even though solar has a low capacity factor at 55°N, its complementarity with wind is essential for the system.

Plot 11 — Onshore Wind Potential Sensitivity

Denmark - Onshore Wind Potential Sensitivity



Modelling Conditions

- Onshore wind p_nom_max reduced from 100% to 0%
- Offshore wind and solar remain at maximum

- Zero CO₂ constraint active

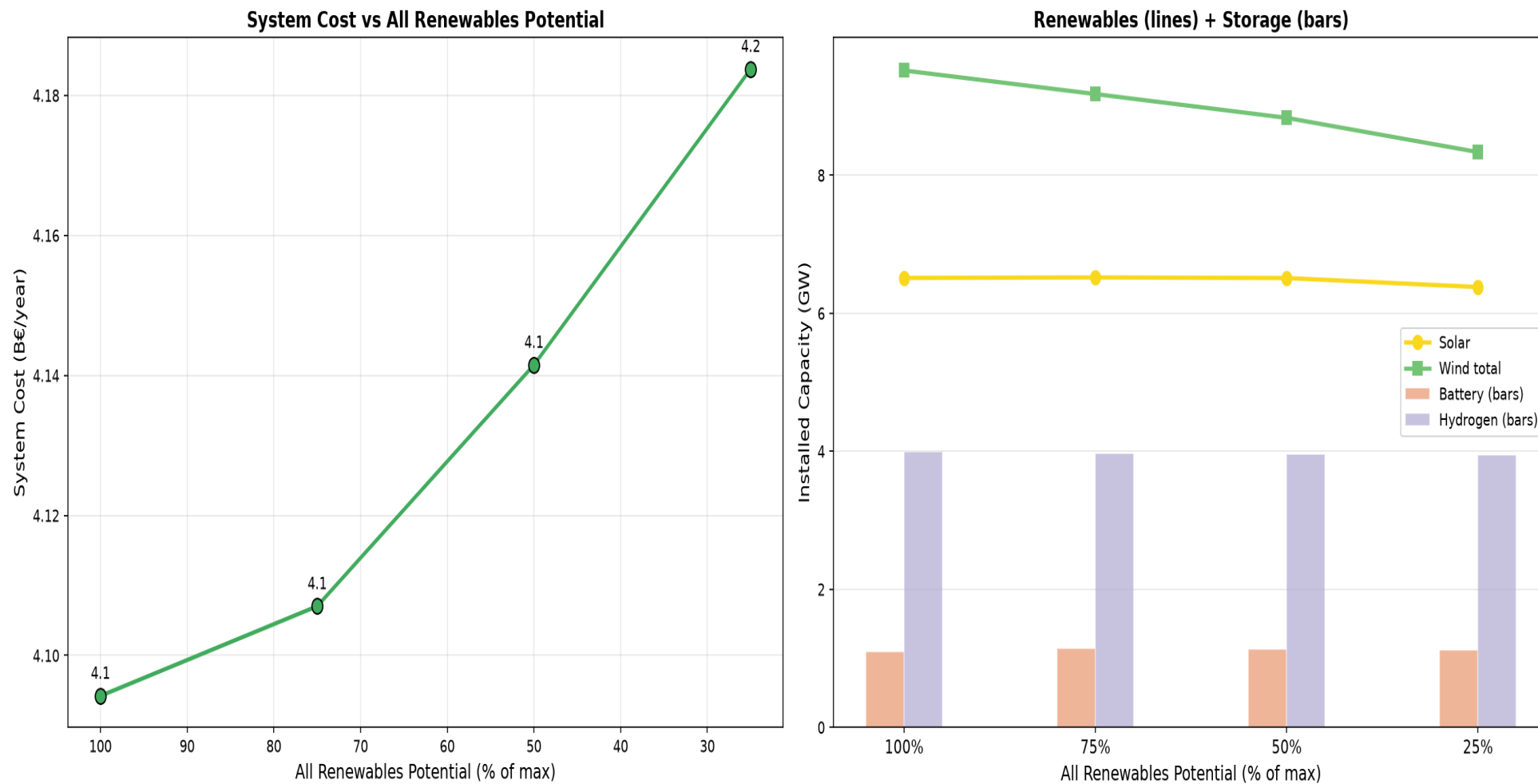
Interpretation & Hypotheses

H1	Restricting onshore wind forces more offshore wind and solar deployment as substitutes.
H2	With the updated larger potentials, the system can now compensate for onshore restrictions — unlike previously when all technologies were at their limits simultaneously.
H3	Offshore wind is the natural substitute for onshore wind in Denmark, given the enormous North Sea potential.

Key Finding: With correct (larger) potentials, onshore wind restrictions are no longer immediately infeasible. Offshore wind can substitute effectively, though at higher cost.

Plot 12 — All Renewables Potential Sensitivity

Denmark - All Renewables Potential Sensitivity



Modelling Conditions

- Solar, onshore and offshore wind all reduced simultaneously
- Worst-case land scarcity scenario

- Zero CO₂ constraint active

Interpretation & Hypotheses

H1	When all renewables are restricted together, costs rise sharply as no substitution is possible between technologies.
H2	Storage (batteries and hydrogen) increases dramatically to time-shift the scarce renewable generation.
H3	This scenario illustrates the importance of proactive spatial planning — protecting renewable energy zones is critical for affordable decarbonisation.

Key Finding: Spatial planning for renewable energy is as important as technology cost reductions. A coordinated national strategy for wind and solar zones is essential.

Plot 13 — CO₂ Decarbonisation Pathway

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Modelling Conditions

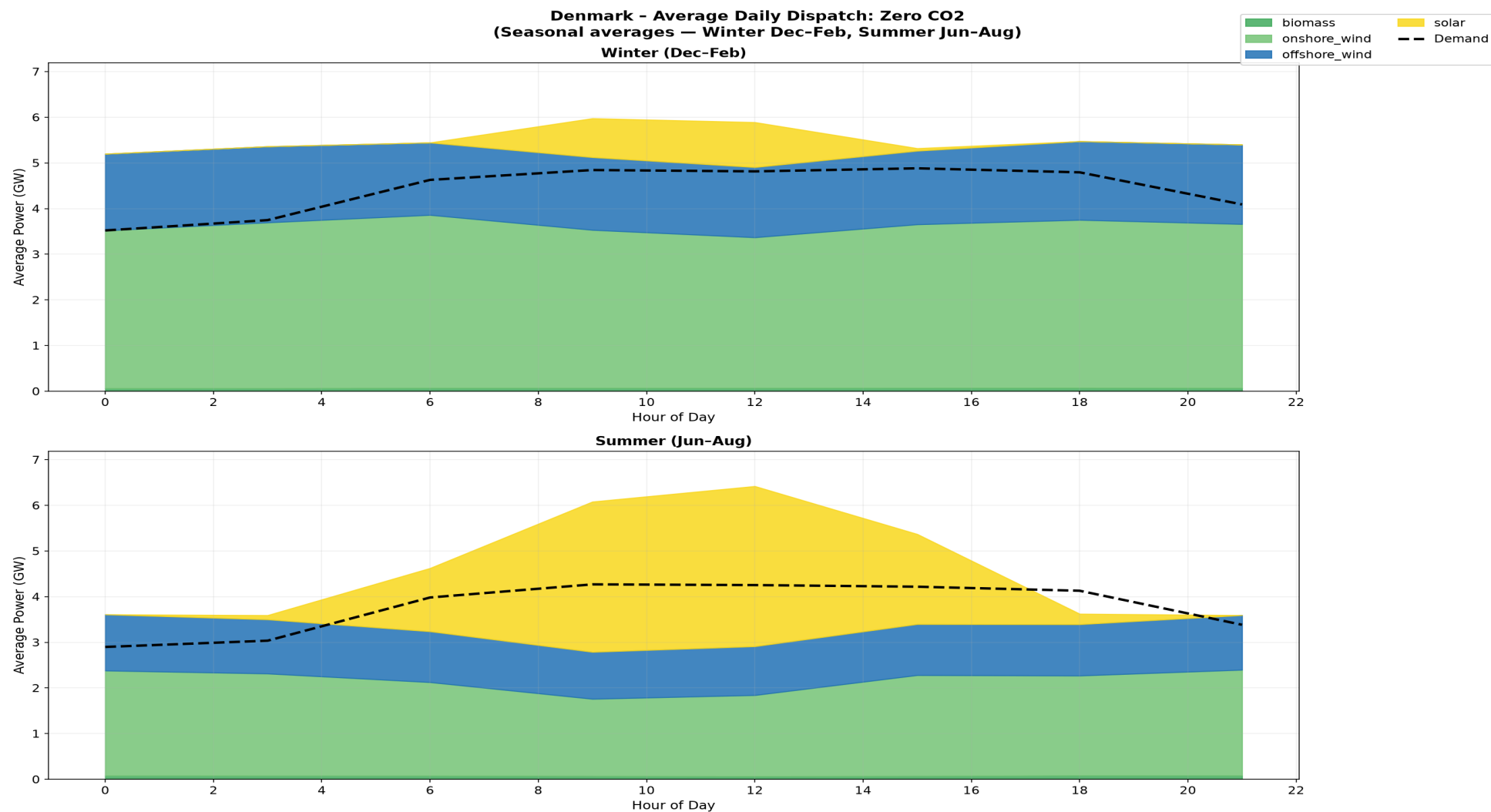
- CO₂ reduced in steps: 0%, 20%, 40%, 60%, 80%, 95%, 100%
- Existing fossil fleet included; CO₂ limit as global constraint
- Full grid expansion, all technologies available

Interpretation & Hypotheses

H1	The first 60% of CO ₂ reduction is cheap because wind directly replaces coal with minimal storage needs.
H2	The last 5% (95% to 100%) costs more than the first 80% combined — rare winter dark-doldrums require enormous storage for very few hours.
H3	Battery storage stays near zero until 80% reduction, then rises sharply — storage is only needed when all fossil backup is removed.

Key Finding: The law of increasing marginal abatement costs is stark. A pragmatic 80-95% target is dramatically cheaper than 100%, and remaining emissions could be addressed through imports or demand response.

Plot 14a — Dispatch Profile: Zero CO₂ Scenario



Modelling Conditions

- Average daily dispatch profile by season (winter Dec-Feb, summer Jun-Aug)
- Zero CO₂ scenario — no fossil generation

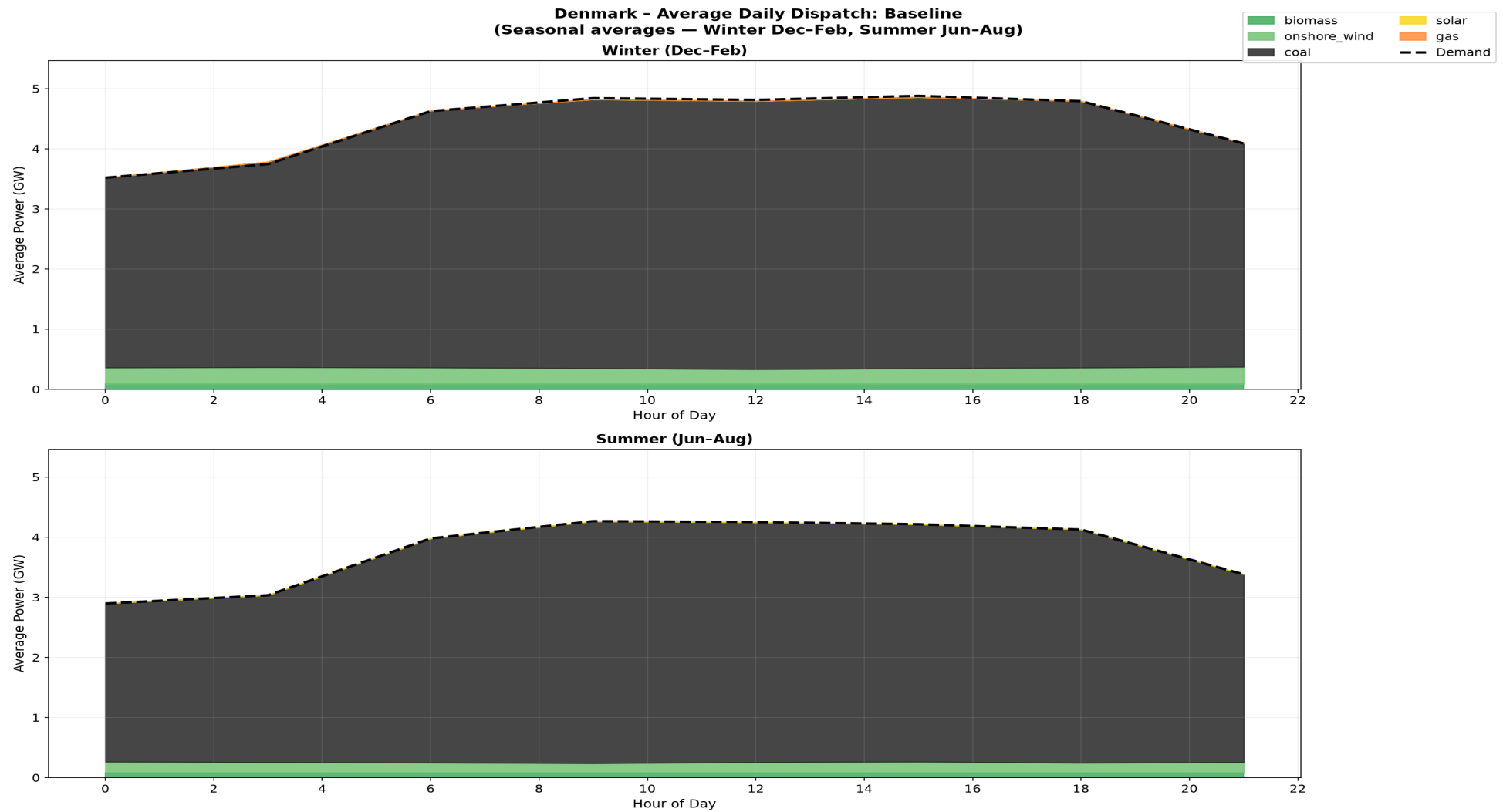
- X-axis: hour of day (0-23); Y-axis: average power in GW

Interpretation & Hypotheses

H1	Winter dispatch is dominated by wind with relatively flat profile — wind does not follow daily patterns like solar.
H2	Summer dispatch shows a clear solar peak at midday, with wind providing baseload and storage smoothing the evening ramp-down.
H3	Demand (dashed line) is relatively constant throughout the day, meaning storage must bridge the gap between variable renewable output and stable demand.

Key Finding: The seasonal dispatch reveals the fundamental challenge: in winter the system is wind-heavy and storage cycles rapidly; in summer solar creates a midday surplus that must be stored for evening and night.

Plot 14b — Dispatch Profile: Baseline Scenario



Modelling Conditions

- Average daily dispatch profile by season
- Baseline scenario — coal provides flat baseload

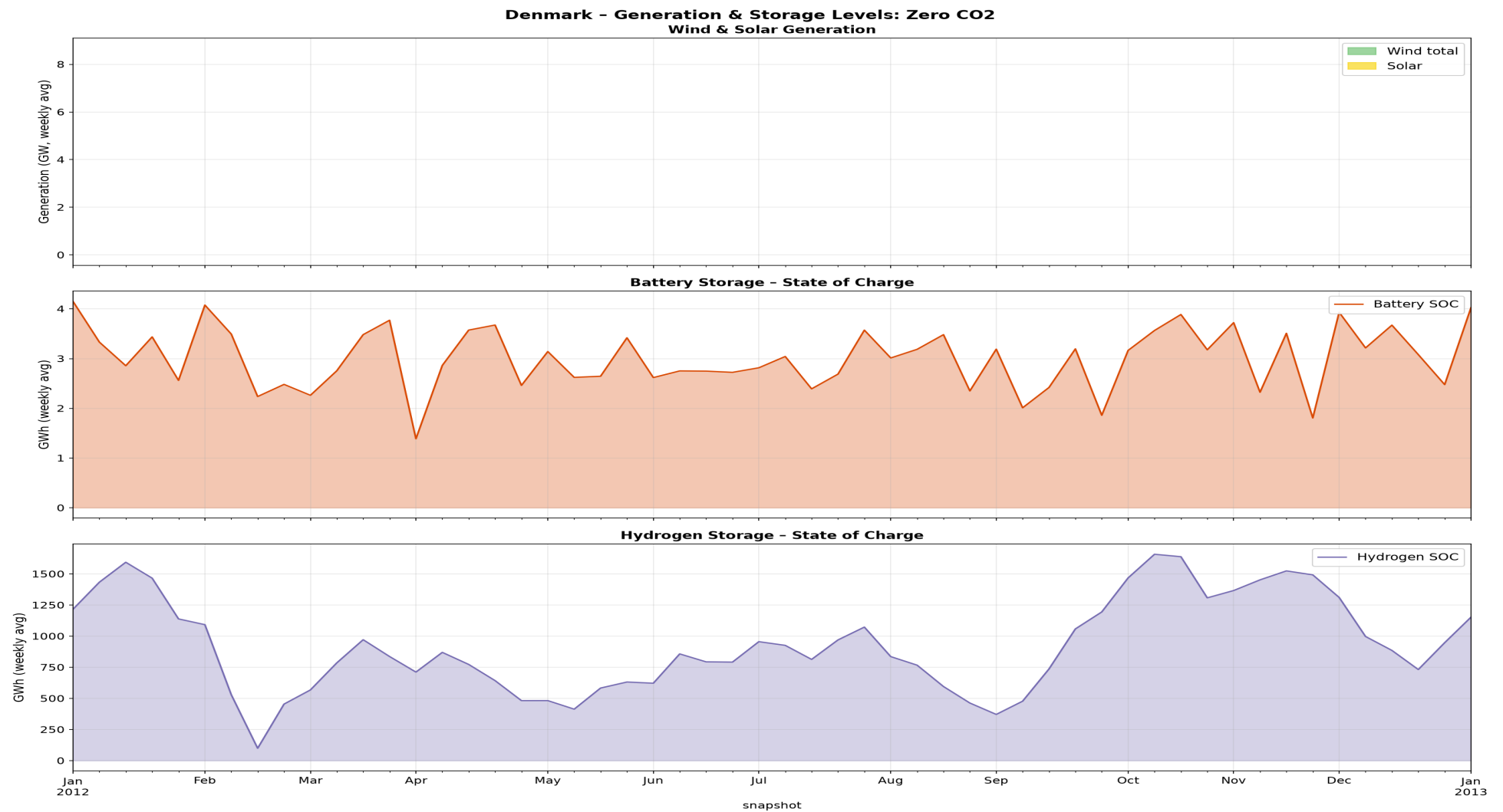
- Wind provides variable supplementary generation

Interpretation & Hypotheses

H1	Coal runs as near-constant baseload in both seasons, with wind providing variable top-up.
H2	Storage is barely visible in the Baseline — coal absorbs all variability, making storage economically unnecessary.
H3	The contrast with Zero CO ₂ dispatch is stark: the Baseline system is simple and predictable; Zero CO ₂ is complex and storage-dependent.

Key Finding: The Baseline dispatch illustrates why coal is so economically attractive in models without carbon pricing — it is perfectly controllable and requires no complementary investments in storage or grid.

Plot 15a — Storage Levels & Generation: Zero CO₂



Modelling Conditions

- Weekly averages of wind/solar generation and storage state-of-charge
- Top: wind and solar generation; Middle: battery SOC; Bottom: hydrogen SOC

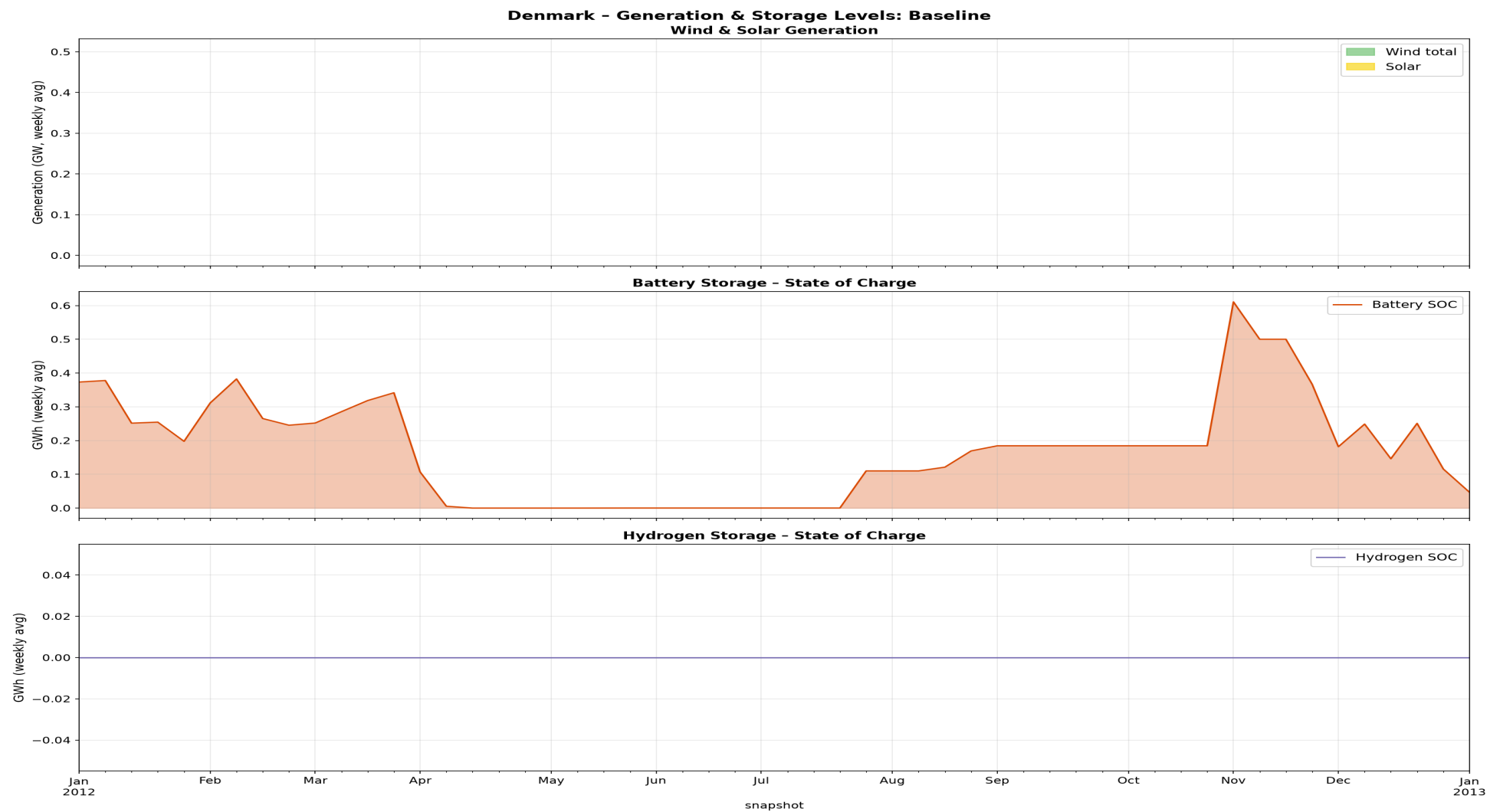
- Zero CO₂ scenario

Interpretation & Hypotheses

H1	Wind peaks in winter (high generation, high battery cycling) and solar peaks in summer — their complementarity reduces storage requirements.
H2	Hydrogen accumulates in spring/summer when renewables exceed demand and discharges in autumn/winter — classic seasonal storage pattern.
H3	Battery cycles rapidly week-to-week following wind patterns, while hydrogen cycles slowly on a seasonal timescale — clear division of labour.

Key Finding: The storage profiles reveal the two-tier storage system: batteries for short-term (days/weeks) balancing and hydrogen for seasonal (months) balancing. Wind and solar generation drives both storage cycles.

Plot 15b — Storage Levels & Generation: Baseline



Modelling Conditions

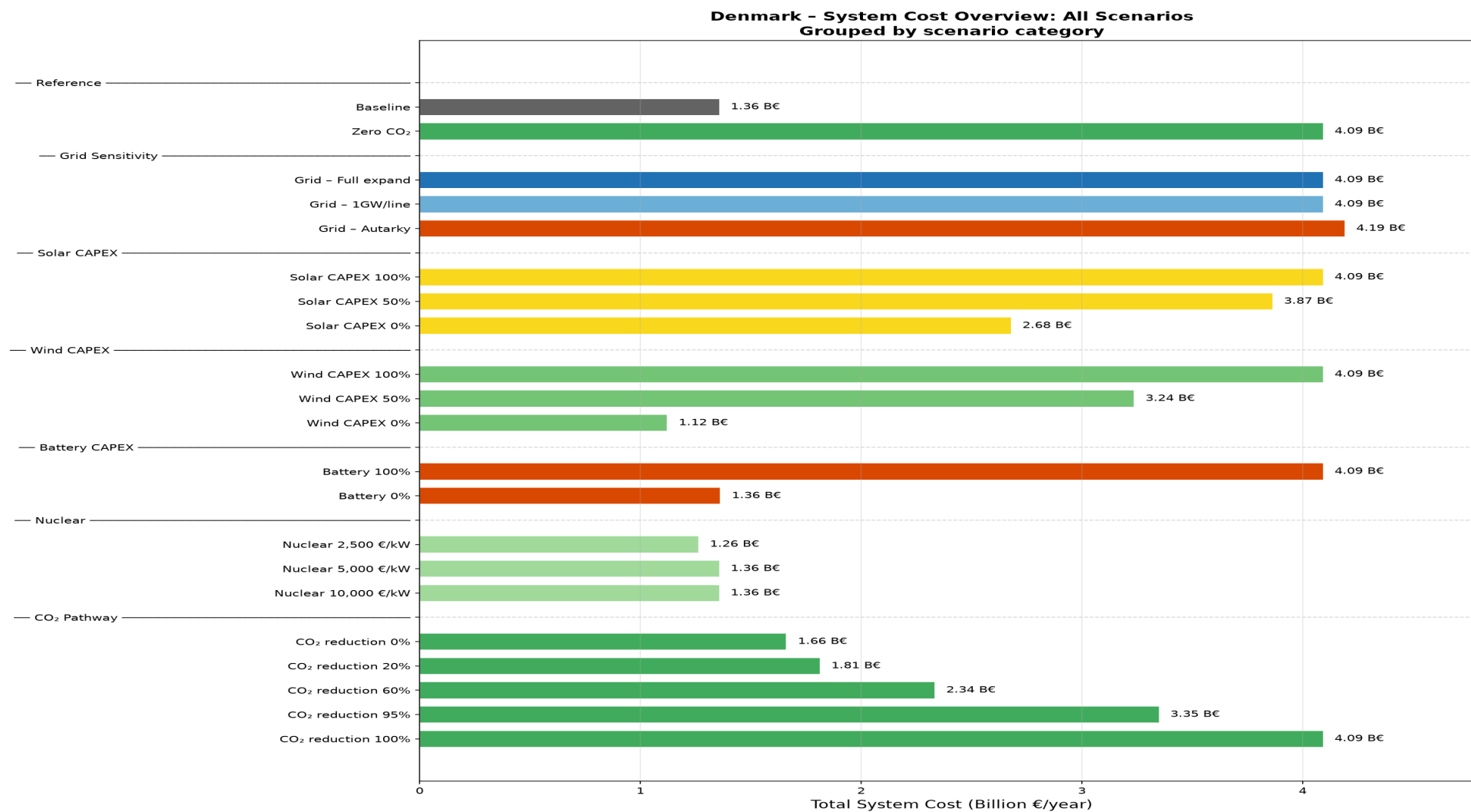
- Baseline scenario — coal provides baseload
- Storage barely used

Interpretation & Hypotheses

H1	Battery activity is minimal and concentrated in winter months when wind occasionally produces surplus.
H2	Hydrogen storage is essentially unused — coal makes seasonal storage completely unnecessary.
H3	The near-zero storage in Baseline vs the large storage in Zero CO ₂ quantifies exactly how much flexibility coal provides.

Key Finding: Coal is a near-perfect substitute for storage. Removing coal transfers its entire balancing function to batteries and hydrogen — at significant additional cost.

Plot 16 — System Cost Overview: All Scenarios



Modelling Conditions

- All solved scenarios compared in descending cost order
- Costs in Billion €/year (annualised investment + operating costs at 7% discount rate)

- Technology costs: 2025 projections

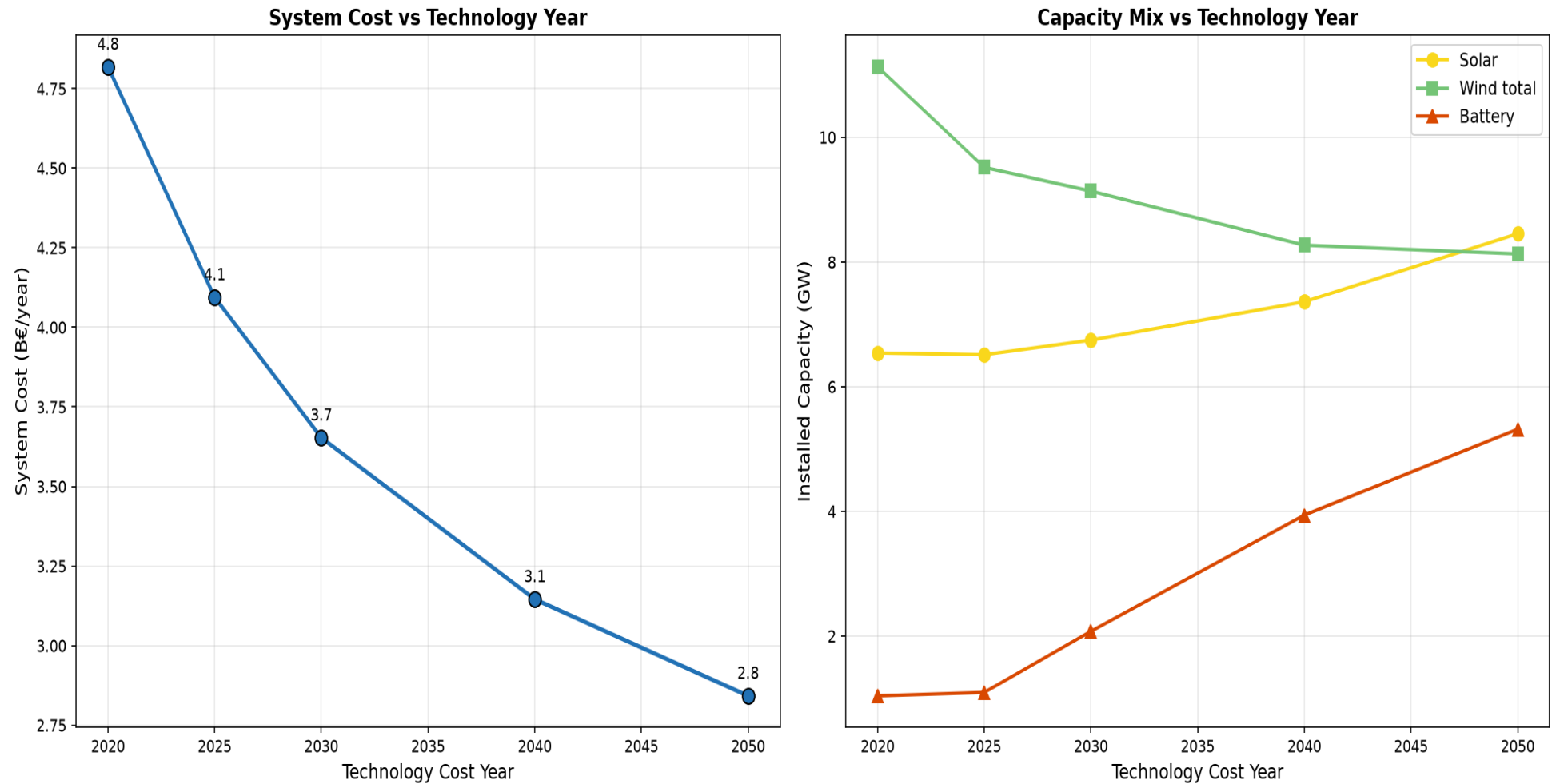
Interpretation & Hypotheses

H1	Autarky is the most expensive real-world scenario — transmission is the most cost-effective flexibility resource available.
H2	Wind CAPEX reduction delivers the largest savings of any sensitivity, reflecting wind's dominant role in Denmark's renewable mix.
H3	CO ₂ pathway scenarios show 60% decarbonisation is affordable; the last 40% costs disproportionately more.

Key Finding: Three key policy recommendations: (1) invest in transmission before local storage; (2) prioritise wind cost reduction; (3) consider a phased 80-95% target supplemented by flexible demand or imports.

Plot 18 — Sensitivity 6: Technology Cost Year Variations

Denmark - Sensitivity 6: Technology Cost Year Variations



Modelling Conditions

- Technology costs from 2020, 2025, 2030, 2040, 2050 projections (PyPSA/technology-data)
- Zero CO₂ constraint active

- Weather year 2012, 3h resolution throughout

Interpretation & Hypotheses

H1	System costs decrease significantly from 2020 to 2050 as all technologies — wind, solar, batteries — become cheaper over time.
H2	Renewable capacity increases in later years as falling costs make it optimal to overbuild rather than invest in storage.
H3	Battery storage increases in later years as falling battery costs make it economical to deploy more short-term storage.

Key Finding: Technology learning curves substantially reduce the cost of zero-carbon electricity over time. The 2050 system is dramatically cheaper than 2020, highlighting the importance of continued R&D investment.

Plot 19 — Sensitivity 7: Temporal Resolution

[Plot not found: 19_sensitivity_time_resolution.png]

Modelling Conditions

- Model solved at 1h, 2h, 3h, 6h temporal resolution
- Zero CO₂ constraint active
- Same weather year 2012 and technology costs throughout

Interpretation & Hypotheses

H1	System costs are relatively stable across resolutions — the model captures the same fundamental trade-offs at all temporal scales.
H2	Storage requirements may decrease at coarser resolutions (6h) as sub-hourly variability is averaged out.
H3	The 3h resolution used throughout this study is a reasonable compromise between computational cost and model accuracy.

Key Finding: Temporal resolution has limited impact on system costs for this model, validating the choice of 3h resolution. Finer resolution (1h) captures more variability but increases computation time significantly.

Overall Conclusions & Policy Implications

Key Finding	Policy Implication
Wind is Denmark's key technology	Prioritise offshore wind cost reduction and protect North Sea wind zones
Storage is the main cost driver	Battery R&D investment is critical — halving battery costs halves system costs
Transmission replaces storage	Grid buildout delivers better value than local autarky — regional integration is essential
Last 5% CO ₂ reduction very expensive	A 95% target plus imports or demand response may be more cost-effective than 100% local
Solar is systemically critical	Protect solar area — complementarity with wind reduces storage requirements
Nuclear only competitive below ~5,000 €/kW	Current European projects not economically justified without a CO ₂ price
Technology costs fall significantly by 2050	Continued R&D investment accelerates the energy transition — timing matters
3h resolution is sufficient	Model results are robust to temporal resolution — computational efficiency validated

Model: PyPSA | Solver: Gurobi 13 | Weather: ERA5 2012 | Eligibility: Copernicus Land Cover, WDPA, GEBCO, Marine Regions | Technology costs: PyPSA/technology-data 2025